

Optimization Methods: Lecture 1

HSE University, FCS

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A.D. Ignatov

aignatov@hse.ru

https://t.me/and_ignatov

General questions

Evaluation

- $O_{total} = 0.6 \cdot O_{sem} + 0.4 \cdot O_{exam}$
- $O_{sem} = 0.28 \cdot O_{colloc} + \sum_1^4 0.18 \cdot O_{hw_i}$

General questions

- How can I avoid doing exam?
 - If $O_{sem} > 8$, considering that $O_{colloc} \geq 7$.

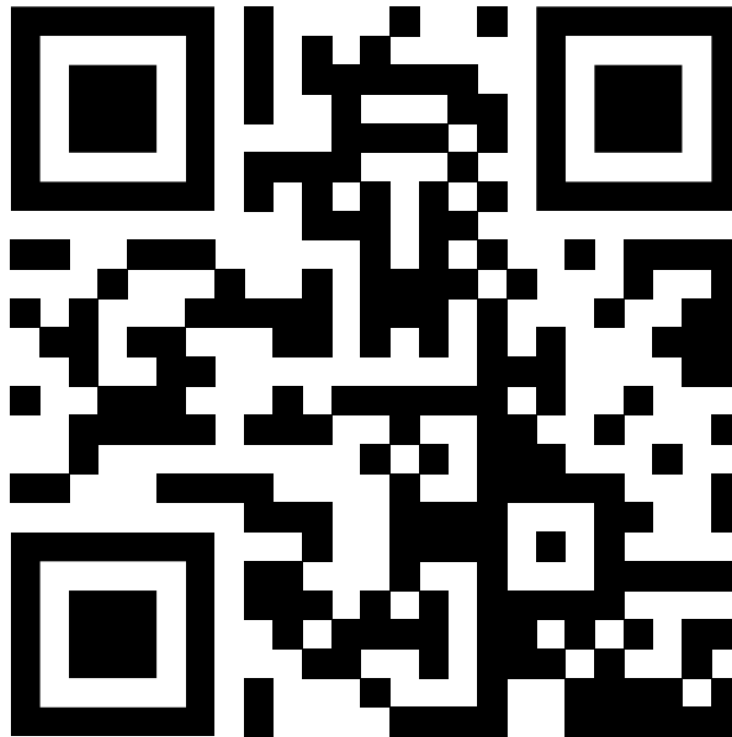
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 - Sure, the Lecturer appreciates it 😊

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- Any other questions?

Let's keep in touch!



Lecture 1. Basic concepts of optimization

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Why do we need this course?

Optimizations Methods: where to use?

- Data analysis
- Machine learning
- **Business analytics**
- Engineering
- Bioinformatics
- ...

Lecture 1. Basic concepts of optimization

Concepts themselves

Classification of Minima

- Let there be a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, defined in some region $\Omega \in \mathbb{R}^n$.
 - A point $x^* \in \Omega$ is a **local minimum** of the function f in the region Ω , if there exists $\epsilon > 0$ such that $f(x) \geq f(x^*)$ for any $x: x \in \Omega \setminus \{x^*\}; \|x - x^*\| < \epsilon$.
 - A point $x^* \in \Omega$ is a **global minimum** of the function f in the region Ω , if $f(x) \geq f(x^*)$ for any $x: x \in \Omega \setminus \{x^*\}$.
 - If we replace the inequalities ' $\geq$ ' with ' $>$ ' in the definitions, we obtain the definitions of the **strict local** and **global minimum**.

Necessary Theory

- Gradient of the function $f: \mathbb{R}^n \rightarrow \mathbb{R}$:

- $\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T$

- Hessian matrix of the function $f: \mathbb{R}^n \rightarrow \mathbb{R}$:

- $F(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_1 x_n} \\ \dots & \dots & \dots \\ \frac{\partial^2 f}{\partial x_n x_1} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$

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Exercise:

Find the determinant of the Hessian matrix of the function $f(x) = 3x_1^2 + 2x_1x_2^2 + 4x_2^2$ at the point $x = (1, 1)$.

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- A convex set Ω : such that for any two points $x_1, x_2 \in \Omega, x_1 \neq x_2$ any point of the line segment $x_1 x_2$ lies in Ω .
- A function that is convex on a convex set Ω : such that for any $x, y \in \Omega, \forall t \in [0,1]$:

$$f(t \cdot x + (1 - t) \cdot y) \leq t \cdot f(x) + (1 - t) \cdot f(y)$$

Feasible Direction

- Feasible direction:

- A vector $d \in \mathbb{R}^n, d \neq 0$ is called a feasible direction at the point $x \in \Omega$, if there exists $a_0 > 0$ such that $x + a \cdot d \in \Omega$ for any $a \in [0, a_0]$.

- Directional derivative:

- Let there be a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$; d is a feasible direction from the point $x \in \Omega$. Then the derivative of the function f in the direction d is defined as

$$\frac{\partial f}{\partial d}(x) = \lim_{a \rightarrow 0} \frac{f(x + ad) - f(x)}{a}.$$

- If $\|d\| = 1$, then $\frac{\partial f}{\partial d}$ characterizes the rate of increase of the function f in the direction d .
- $\frac{\partial f}{\partial d}(x) = d^T \nabla f(x)$.

First-Order Necessary Condition

- Let the function $f \in C^1$ be defined on the region $\Omega \in \mathbb{R}^n$. If x^* is a local minimum of f on Ω , then for any feasible direction d at the point x^*
 $d^T \nabla f(x^*) \geq 0$.
- If x^* is an interior point of the set Ω and it is a local minimum of f on Ω , then $\nabla f(x^*) = 0$.

Second-Order Necessary Condition

- Let the function $f \in C^2$ be defined on the region $\Omega \in \mathbb{R}^n$; x^* is a local minimum of f on Ω ; d is a feasible direction at x^* . If $d^T \nabla f(x^*) = 0$, then $d^T F(x^*)d \geq 0$.
 - If x^* is an interior point of the set Ω and it is a local minimum of f on Ω , then $\nabla f(x^*) = 0$ and $d^T F(x^*)d \geq 0$ for any direction d .
- It is important to understand that satisfying one or both necessary conditions does not guarantee a local minimum at the point.

Second-Order Sufficient Condition

- Let the function $f \in C^2$ be defined on the region $\Omega \in \mathbb{R}^n$; x^* is an interior point of Ω ; $\nabla f(x^*) = 0$; $F(x^*) > 0$. Then x^* is a strict local minimum of f .